Compound Gear Train Worksheet

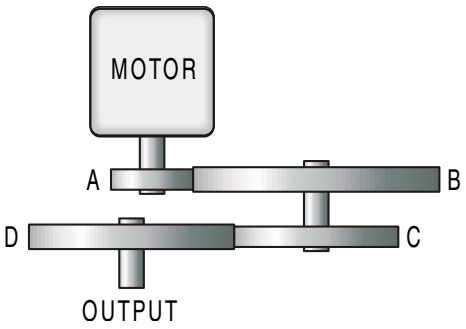
Q1. A compound gear train is shown below and is driven by a motor that runs at 1000 revs min^{-1}.

A = 20 teeth

B = 60 teeth

C = 40 teeth

D = 50 teeth



Calculate;

(a) The velocity ratio of the system.

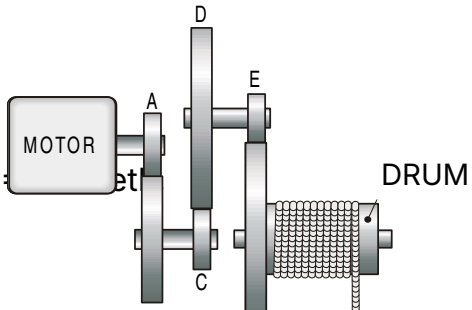
Substitute both velocity ratios into the original formula,

(b) The output speed.

Q2. A motorised winch is shown opposite.

A = ?

B = 32 teeth, C = 15 teeth, D = [illegible]



E = 12 teeth, F = 48 teeth

If the motor runs at a speed of 1200 revs min^{-1} and the drum is to rotate at 25 revs min^{-1}, calculate;

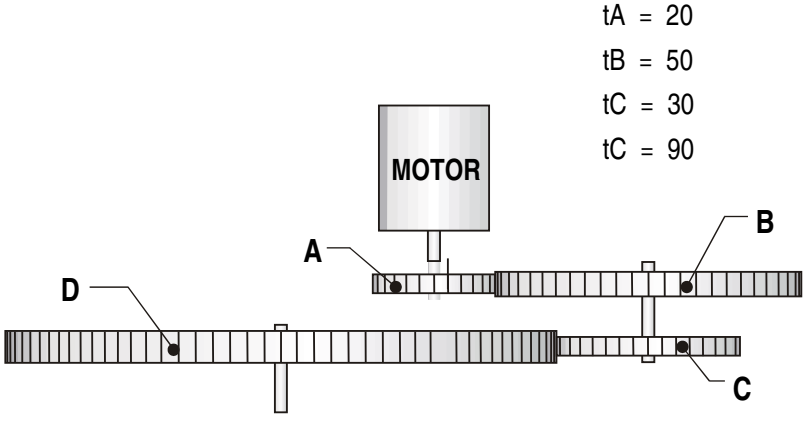
(a) The velocity ratio required to produce this speed reduction.

(b) The number of teeth gear A must have to meet this requirement.

Substitute velocity ratios into the original formula,

3. The compound gear train shown is driven by a motor that runs at 750 revs min^{-1}. Calculate;

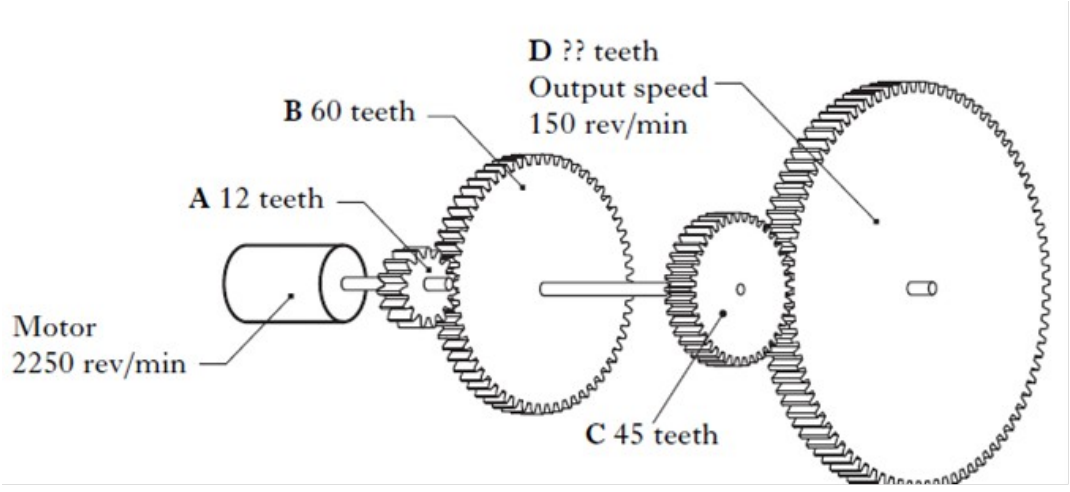
(a) The velocity ratio.



Substitute velocity ratios into the original formula,

(b) The output shaft speed.

Q4. A gear system for operating a theatre curtain is shown.



(a) State the name of the type of gear system.

Compound gear train

(b) Calculate:

(i) The velocity ratio of the system when gear D rotates at 150 revs min^{-1}.

(ii) The ratio of gear B to gear A;

(iii) The number of teeth on gear D to produce the correct output speed.

(c) State **one** advantage of a gear system made from plastic rather than steel gears.

Self-lubricating (less friction), less mass, cheaper

Q5. A winch is to be designed for the front of an off-road vehicle.

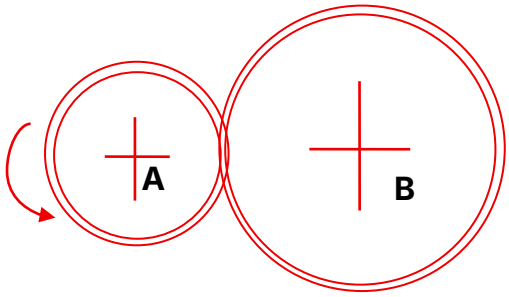


Use ***Yenka*** to simulate:

a) A simple gear train where the output turns 5 times slower than the input. (Use a constant speed motor as an input). Draw your solution using the correct symbols for this simple gear train.

Input gear A must have 0.2 (one fifth, 1/5) the number of teeth as output gear B.

Put another way, B must have 5 times the number of teeth of A.



b) A compound gear train where the output turns 20 times slower than the input. (Use a constant speed motor as an input). Draw your solution using the correct symbols for this compound gear train.